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Dynamic grid stability in low carbon power systems with minimum inertia

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ABSTRACT

The power system transition to large penetrations of renewable generation has become a core target of decarbonisation roadmaps in many countries. However, switching from synchronous to variable inverter-based resources has prompted technical concerns linked to the reduction in system inertia and short circuit levels. This may result in significant deviations in frequency and bus voltages during fault events. Hence, this paper provides a comprehensive review of power system stability challenges when integrating up to 100% inverter-based resources. The potential solutions from several enabling technologies (i.e., energy storage, controllable loads, wind, solar PV, etc.) are also assessed to mitigate anticipated issues. The review further details the role of grid codes and international standards in maintaining dynamic stability in power systems with extremely high up to 100% variable renewable generation. Finally, discussion and recommendation pathways are provided to inform industries and accelerate a secure power system transition to meet Net Zero targets.

1. Introduction

The 2015 Paris Agreement provides long-term goals to reduce greenhouse gas (GHG) emissions to limit the global temperature increase to less than 2° Celsius, a mechanism to review country emissions every five years and finance to developing countries to prepare for climate change [1]. In essence, it initiated the beginning of the global Net Zero transition. The findings of the 2018 Intergovernmental Panel on Climate Change (IPCC) were clear: global carbon-neutrality is required by 2055 if we are to achieve the Paris Agreement goals and prevent catastrophic runaway global warming [1,2]. When compared to transport and heating, power systems have decarbonised the fastest [3].

The United Kingdom (UK), Australia, the European Union (EU), the United States of America (USA), and Brazil have set a Net Zero GHG emissions target by 2050, and it is expected that these countries will transition to a nearly 100% renewable electricity system [4]. Falling wind and solar photovoltaic (PV) prices have been the key driver in accelerating this transition [5]. Global wind and solar PV installations have grown rapidly [6], as shown in Fig. 1. In fact, wind and solar power installations grew significantly from 220 GW to 73 GW in 2011 to approximately 733 GW and 713 GW, respectively in 2020 [6]. Denmark, Ireland and Portugal have already reported periods of nearly 100% instantaneous renewable power for a few hours [7].

However, managing a power system with 100% renewable generation is fundamentally different from operating a partially renewable power system. Wind and solar power are not without their challenges, mostly related to the stochastic and intermittent nature of renewable resources [8,9]. Energy storage systems are playing a role in this transition to ensure system flexibility and reliability, but they are not a silver bullet solution [10]. Therefore, switching from synchronous fossil fuel-based generation power systems to variable inverter-based renewable resources raises a number of technical concerns.

The first key challenge is the intermittency and variability of renewable generation due to their weather dependency, which has prompted questions regarding the reliability and flexibility of a power network [11]. Given the fluctuation and uncertainty of wind and solar PV, system operators must precisely estimate future natural resource availability as well as extreme weather conditions [12]. In this case, improved forecast accuracy plays a critical role in reducing the uncertainty connected with natural resources.

However, due to the impact of weather conditions on both generation and demand, all forecasting approaches incorporate stochastic error [13]. Lowering prediction error would assist grid operators in forecasting longer low wind occurrences, maintaining the long-term stability and security of renewable production [14], and reducing the reserve capacity requirements [15]. This is because estimating reserve is

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dependent on variable renewable energy (VRE) generation, and the unpredictability associated with loads.

The second challenge is that many of these new resources are connected to the grid via power electronic inverters rather than by spinning electromechanical machinery [16]. When compared to synchronous generators, inverter-based resources have very distinct characteristics, including a lack of physically coupled inertial response and, traditionally, a limited ability to produce a high short circuit current during fault events [17]. Therefore, from a dynamic stability point of view, shifting from a dispatchable centralised power system with inherent self-synchronisation to inertia-less inverter-based resources (IBR) means gradually losing system inertia and control mechanisms. Consequently, this will change the dynamic behaviour of the power system.

All these issues have resulted in a lack of frequency and voltage management, network congestion, and a deterioration of system adequacy and restoration capabilities [17]. This issue is particularly apparent in islanded grids with few synchronous interconnections, such as Ireland, the UK, and Australia, and even in parts of the USA and China which are not synchronously interconnected to larger power systems.

According to several studies, there are sufficient renewable resources around the world to achieve 100% renewable electricity [18–20] by 2050. These concepts, however, have been criticised for relying heavily on assumptions and energy modelling rather than dynamic grid simulations to evaluate if a power system with a 100% VRE system is viable [21,22]. A number of reviews evaluate the impact of small to medium shares of VRE on power system operation [23,24].

However, the main concern that yet to be addressed is what is the impact of extremely high VRE, nearly 100% VRE, on the power system dynamic stability and what are the potential flexibility solutions to securely operate the future power system. Hence, the main objective of this paper is to provide a comprehensive review of the technical challenges and barriers to power systems with minimum inertia. This review is then expanded to examine potential solutions and challenges with currently available technologies to improve frequency and voltage control in future power systems.

This rest of this paper is organized as follows: Section 2 provides an overview of the dynamic stability challenges that affect grid operations. Section 3 describes how these challenges have been addressed to date. Section 4 discusses the role of grid codes and renewable interconnection standards. Section 5 addresses the grid codes requirements for distributed generation resources. Section 7 concludes with a consideration of the future 100% VRE roadmap after a brief discussion in section 6.

2. Power system dynamics with increasing renewable penetration

Power system stability is defined as the ability of an electrical power system to maintain stable operation after being subjected to large fault events. There are three types of stability associated with the power system: rotor angle stability, voltage stability, and frequency stability [23,25]. All three must be met all the time to maintain the security of the network. In the following subsections, the voltage and frequency stabilities are explained while the rotor angle stability is out of scop of this study.

2.1. Frequency stability

The dynamic power balance between generation and load is met by controlling system frequency within specified operating limits around a nominal frequency. When demand exceeds generation, the system frequency declines, and vice versa, thus frequency is a power balancing metric between generation and demand [26,27]. The spinning masses of the synchronous generators coupled with the grid provide a natural inertia response to unforeseen disturbance if the frequency dips [23].

In an AC system, the inertia response is the most significant feature for reducing the speed drops of rotating machines as well as the system frequency. It buys time for governor-equipped synchronous generators to operate and arrest frequency changes in the early stages of a disturbance and is frequently referred to as primary frequency response. This is followed by the secondary frequency response, automated generation control, which restores the system frequency to normal operation [28,29]. Fig. 2 illustrates the main steps of frequency recovery of a conventional power system following a large disturbance [11]. It can be seen that the rate of change of frequency (RoCoF) and frequency nadir (i.e., angle) are inversely proportional to the system inertia. In power systems with low levels of synchronous generation, there are low levels of system inertia which results in higher RoCoF and lower frequency nadir.

There is a direct relationship between the generator's moment of inertia and kinetic energy stored in the generator's rotating mass. The kinetic energy of every single generator can be expressed as

$$KE = \frac{1}{2}\, J.V^2 \tag{1}$$

where *KE is* kinetic energy and measured in (watt-second), *J* represents moment of ineria in (Joules), and *V* depicts angular velocity in (rad/second) [27].

Generator inertia constant (*H*) can be derived from equation (1), which is equal to kinetic energy (MW s) at generator rated speed over

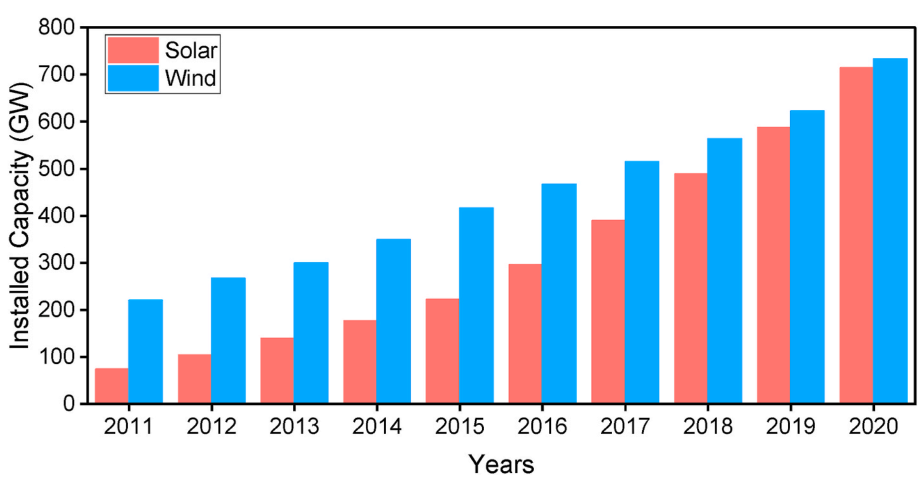

Fig. 1. Global installed wind and solar capacity between 2011 and 2020 [6].

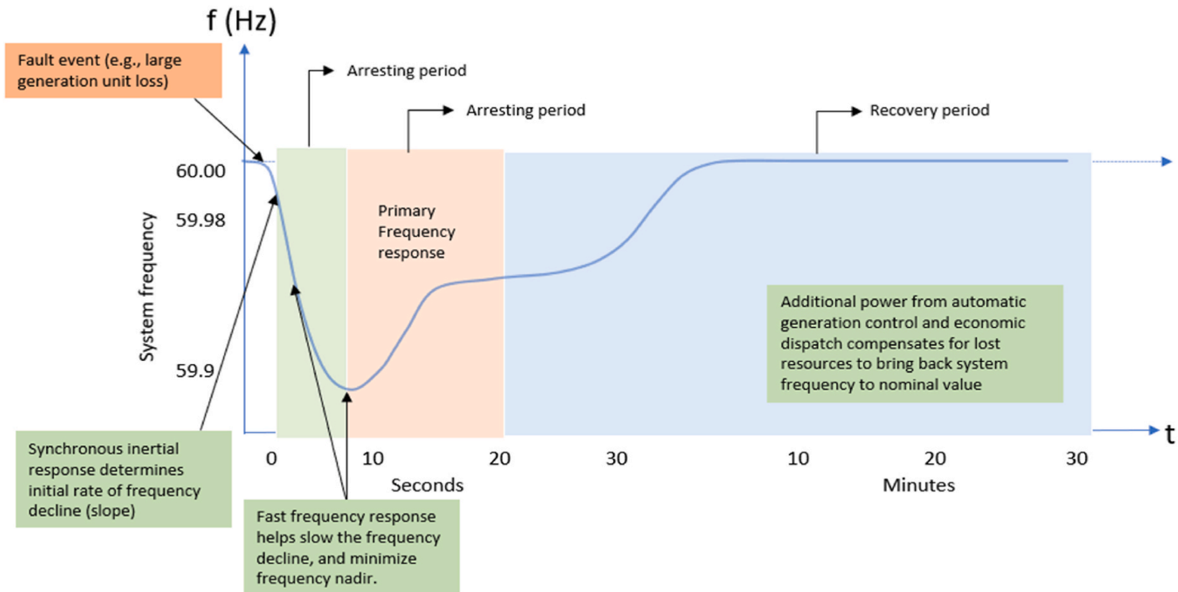

Fig. 2. Typical frequency control systems following generation loss [23].

generator rated power (S_{VA}) in (MVA) as expressed in equation (2).

$$H=\frac{1}{2}\frac{JV^2}{S_{VA}}=\frac{KE}{S_{VA}} \tag{2}$$

It is worth mentioning that the inertia constant is different from one generator to another depending on generator rated power, type, and size. Therefore, the net inertia (H_{sys}) of any interconnected system can be derived as follows:

$$H_{SYS}=\frac{\sum_{i=1}^{n}KE_i}{S_{sys}}=\frac{\sum_{i=1}^{n}H_i\,S_{VA_i}}{S_{sys}} \tag{3}$$

where H_{SYS} and S_{sys} are the system net inertia and net rated power, KE_i, H_i, and S_{VA_i} are the stored rotating energy, inertia constant, and rated power of the i-th generator, and n is the number of online generators.

Similarly, the relation between system inertia and RoCoF can be derived from the swing equation as follows:

$$\frac{df}{dt}=\frac{f_0}{2\,H_{sys}S_{sys}}\,(P_m-P_e)=\frac{f_0\Delta P}{2\,H_{sys}S_{sys}} \tag{4}$$

where $\frac{df}{dt}=\Delta f$ is RoCoF, $P_m-P_e=\Delta P$ are the mechanical power and electrical power in (MW), respectively, and f_0 is the system nominal frequency in (Hz).

This allows us to define the governor droop setting for each generator R_i as in equation (5). R_i represents the rate at which the i-th synchronous generator is expected to inject/absorb power in response to frequency deviation when the system is subjected to contingency.

$$R_i=\frac{\frac{\Delta f}{f_0}}{\frac{\Delta P_i}{S_{VA_i}}},\ i=1,\dots,n \tag{5}$$

As VRE grows, so does the retirement of conventional synchronous machines, resulting in a significant drop in system inertia, which has a detrimental impact on the frequency nadir and RoCoF [17,30]. This is a key challenge with higher and higher penetrations of VRE. This is simply because VRE is electronically decoupled from the grid and cannot provide a natural response to system events. A faster acting reserve is required in response to increased VRE.

A low inertia grid requires a faster frequency response after abnormal events [31]. In extremely low inertia systems, frequency variations can cause cascade tripping of generation units, under frequency load shedding, and, in the worst-case situation, blackouts [32]. Hence, as a result, new technologies are urgently required to bridge the gap between synchronous inertia and primary operational reserve in a rich VRE power system. However, fast response services from VRE can effectively contribute to a reduction in the power system frequency nadir and improve RoCoF during the fault events if the VRE has the correct converter control technologies.

Many countries have introduced new frequency services to meet inertia requirements with increasing shares of VRE penetration. For example, the transmission system operator (TSO) of Northern Ireland (SONI) and the Republic of Ireland (EirGrid) have introduced a new program called Delivering a Secure, Sustainable Electricity System (DS3) to supply new fast frequency response (FFR) services is a response to extreme events [33]. The objective of this FFR service is to increase MW output from either inverter-based resources or to reduce load demand in ≤ 2 s after initiation of a frequency event for a duration of 8 s [33]. Although the FFR service with the current available technologies may not be able to response fast enough to effectively act on maximum RoCoF, the service can reduce the frequency deviation and hence supports a primary frequency response.

The UK National Grid (NGUK) introduced four new faster acting frequency response services called dynamic regulation, dynamic moderation, dynamic containment, and static containment to obtain an even faster frequency response within 0.5 s after an event is initiated [34]. These services are beneficial in reducing RoCoF severity and giving other sources of reserve time to respond to frequency excursions. Dynamic regulation is a slow dynamic response that is used for maintaining frequency within a nominal value. Dynamic moderation is a fast dynamic response to keep frequency within operational limits. The main difference between these two services is the frequency dead band.

Dynamic and static containments are designed for post fault operation (i.e., for deployment after a large frequency deviation) to meet the need for faster acting response immediately. The motivation for introducing these services in the UK is to enable 100% active power from storage in less than 1 s after a frequency deviation is initiated [35]. Thus, maintaining a grid frequency close to 50 Hz of normal operation. Other countries have also introduced similar FFR services [23]. Some of these FFR services are summarised in Table 1.

Although these FFR services can deal with low system inertia and high RoCoF challenges they cannot guarantee grid stability in zero inertia systems. New services with much faster acting frequency

Table 1
International new fast frequency response service.

Country	TSO	System services	Response time second	Duration second	Service trigger
UK	National Grid	Dynamic regulation	0.5	1200	Post-fault contingency
		Dynamic moderation	2	infinite	
		Dynamic containment	0.5	1200	
		Static containment	1	1800	
Ireland	EirGrid	FFR	≤2	8–10	Post-fault contingency
Northern Ireland	SONI				
USA	ERCOT	FFR	0.25	900	Post-fault contingency
		LR	0.5	3600	
	PJM	FFR	2	Sustained	Post-fault recovery
Australia	AMEO	FFR	0.1–1	6	Post-fault contingency
Canada	HQT	FFR	1.5	9	Post-fault contingency
	AESO	FFR	0.3	N/A	
Nordic	SOA	FFR	0.7–1.3	5–30	Post-fault contingency
New Zealand	Electricity Authority	FIE	1	60	Post-fault contingency
		SIR	60	900	Post-fault recovery

response need to be introduced in the future when synchronous generators are completely phased out. This is because active power response with a shorter time delay has a significant impact on the system frequency nadir and hence RoCoF [36].

According to the study [35], frequency nadir reduces linearly for fast activation of FFR with a delay time of less than 0.5 s. In low inertia power systems, this faster response is vital to limit frequency nadir when the disturbance is initiated. Active frequency response with less than 0.15 s and 0.3 s trajectory ramp time can be obtained from BESS [36] or supercapacitors [37]. Achieving such a fast-acting response can significantly improve the frequency nadir and RoCoF of low inertia grids.

Another approach that may be used to manage higher and higher VRE levels is to introduce changes in the definition of the system inertia and RoCoF in a national grid code. For example, the TSOs in Ireland and Northern Ireland have a target to move the maximum power system RoCoF operating limit from ±0.5 to ±1 Hz/s to facilitate the secure operation of the system at 75% inertia-less renewable generation [38]. This approach is moving the goalposts of acceptable system operation (accepting a higher level of dynamics) rather than fully compensating for reduced inertia in high VRE grids.

Although a large amount of research has been done to investigate the relationship between frequency stability and system inertia [39,40], none of these studies has defined the future power system needs to securely operate with 100% inverter-based VRE systems. The minimum levels of synchronous generation needed to maintain grid stability in power systems with up to 100% VRE are relatively unknown. Therefore, research is urgently needed to estimate the amount of synchronous generation needed in such very high VRE systems to ensure grid stability and service levels while transitioning to Net Zero by 2050.

2.2. Voltage stability

Voltage stability is the ability to regain steady-state voltage after being subjected to a large contingency event, such as a sudden increase in load or grid fault [41]. Loads are not served efficiently if a stable voltage is not appropriately maintained, and this may result in outages and blackouts [42]. As a result, it is critical to keep system voltages across all buses within acceptable limits in order to ensure that secure and reliable power is delivered to the transmission and distribution system. Reactive power balancing, generation, and absorption are commonly employed to maintain voltage stability in power systems. Traditionally, conventional synchronous generators have been the principal source of reactive power balancing.

Voltage stability relies on the reactive power capability of conventional fossil fuel power plants and the reactive power consumption of distributed loads, but it is also affected by the voltage control strategy of the system [30]. System voltages must be kept within operating limits, otherwise violations may lead to power outages and blackouts. The power system must be able to sustain stable voltage and power when sudden changes occur in system loading. Total system blackouts can occur when this is not possible, as was the case with the blackouts that occurred in Japan in 1987, and in the USA, Canada, Sweden, and Denmark in 2003 [17]. Therefore, understanding dynamic voltage stability particularly in power systems with 100% inverter based VRE is crucial.

Currently, the well-known P–V and Q-V curves are commonly used to study voltage stability analysis [23]. Unlike synchronous generators, inverter based VRE cannot inherently absorb/inject required reactive power in response to voltage changes [43]. However, they can be used for voltage regulation purposes if equipped with proper control strategies [42]. It is to be noted that involving inverter based VRE in voltage regulation will reduce their capability to provide active power response [41]. The Q-V control technique and optimization of renewable generators will provide the required voltage support and reduce the negative impact on their ability to produce active power [25]. Battery energy storage is widely used in combination with wind and PV generators to provide reactive power support [44].

One of the key challenges to addressing voltage stability in systems with a high share of VRE generation is the inability of these technologies to ride through a voltage dip after being subjected to disturbances [23]. Photovoltaic and variable speed wind turbines can regulate their reactive power output in response to voltage changes at the point of common coupling (PCC) using modern voltage control systems such as grid forming inverters with sufficient battery energy storage [17]. In contrast, fixed speed wind turbines consume reactive power during disturbances due to acceleration in their turbine speed. Consequently, reactive power compensator technologies (i.e., capacitor banks) are deployed to provide voltage drop support at the PCC to ride through the disturbance condition [45].

The second challenge is that inverter-based technologies can only offer limited short circuit current. Therefore, a transition to a 100% VRE power system means gradually declining short circuit levels (SCL), which will result in lower retained voltage due to the inability of the system to balance reactive power requirements and ride through faults. Reactive power is a key enabler to maintain voltage stability in the system.

3. Enabling technologies solution

Several technologies have been introduced to tackle challenges in power systems with increasing inverter based VREs. These include energy storage, demand response, electric vehicles, synchronous condenser, High voltage direct current (HVDC) interconnectors, enhanced inverter control technology, the role of the electricity market, cyber security. The potential of these technologies to address grids stability issues in a system dominated by intermittent renewable energy is reviewed in the following subsections.

3.1. Energy storage solution

Energy storage is crucial in renewable rich power systems. Energy storage absorbs energy at high peak generation and uses it later at high peak demand. It can be controlled to provide ancillary services, frequency support and voltage regulation, time shift and black start [46]. Energy storage can be scaled from a few kWh up to hundreds of MWh. A detailed discussion of key storage technologies can be found in Ref. [47].

Energy storage technologies can be mechanical storage (e.g., pumped hydro energy storage (PHES), compressed air energy storage (CAES), flywheel energy storage (FES)), chemical (e.g., hydrogen energy storage (HES), thermal (e.g. thermal energy storage (TES) using phase change materials and water in electric storage heaters), electro-chemical (e.g., lithium-ion, lead acid, and flow cell), electrical (e.g., supercapacitors (SCs), and superconducting magnetic energy storage (i.e. SMES)). Table 2 summarises the techno-economic parameters of these energy storages technologies [37,46,48].

Pumped hydro energy storage is a mature technology with the highest universal installed capacity of 168 GW in 2019 as opposed to only 4 GW of other energy storage systems [49]. The technical operating details and the principle of PHES can be found in Ref. [50]. It offers several desirable services such as short and long-term energy management, power back-up, demand lopping, valley filling, and natural response (inertia) [51]. Although PHES are highly effective, they are limited by being restricted to geographically suitable locations. They also have a high installation capital cost and are often subjected to barriers such as environmental sanctions.

Compressed air energy storage works on a similar principle to PHES [52]. It has many advantages over PHES such as less capital cost than PHES (3-5£/kWh), scalable power output, and quick start-up (9–12 min from the black start, faster from spinning reserves) [47]. There are only two grid scale CAES facilities in the world, 110 MW McIntosh in the USA, and 290 MW Huntsorf in Germany [53]. In both plants, salt caverns are used to store compressed air. This technology is associated with two major problems: 1) the air heats up as it is compressed, which reduces the amount of air that can be stored; 2) when the compressed air is released to spin a turbine the pressure in the cavern is reduced [54]. These result in less electricity production. CAES also has geographical restrictions and material import limitations. Therefore, BESS is superior to CAES for short term power balancing and fast active stability service.

Power to gas (i.e., HES) and power to heat (i.e., TES) energy storage can play a key role in buffering the variability in wind and solar generation to accommodate system frequency and voltage at a certain level [55]. In terms of price, they are much cheaper than BESS, while in comparison to the geographical limitations of PHES and CAES, they are more flexible [56]. Therefore, HES and TES have a unique opportunity to be extremely low cost and flexible storage options to enable a 100% renewable energy system under different weather conditions [55]. Flexibility is essential to maintain a power balance between generation and demand.

Traditionally, fossil fuelled synchronous generators are dispatchable and inherently flexible. However, for VRE to truly compete with fossil fuel, HES and TES are the cheapest options for providing flexible supply and demand at multiple time scales. They are considered by some as one of the better storage medium solutions to store electrical energy for a long duration due to their high storage capacity potential [57,58]. However, HES and TES are not mature technologies and are still under development. Efficiency is a major concern for both technologies due to power loss combined with multiple power conversions. Technical details for both technologies can be found in Refs. [59,60] respectively. Furthermore, social impact and associated challenges (i.e., hydrogen leakage for conflagration, explosion of high-pressure tank hydrogen, and self-ignition) need to be considered when deciding on expanding HES [61].

In contrast, due to modularity, scalability and not being restricted to geographical locations, batteries have dominated the electricity storage markets, including the utility service, residential, and EV markets. Battery energy storage systems (BESS) have become more prolific in power systems with increasing levels of VRE generation [62,63]. Due to the much lower cost of batteries, they are a better option for short-term storage (minutes to hours) when compared to PHES, CAES, HES, and TES [50,64]. However, due to the life cycle and capacity limitations, batteries are more effective for large duration energy storage applications (several hours to weeks). Therefore, installing multiple storage options can be considered one of the optimum solutions for providing the range of ancillary services needed to completely replace conventional fossil fuel power plants with VRE generation [65].

3.1.1. Battery energy storage system

Battery energy storage systems (BESS) are portable, modular, and can be installed easily in comparison to PHES and CAES. Battery energy storage systems are commercially available in different sizes ranging from a few kW to grid-scale capacities (>100 MW). Scalability and sizing are the two most important features of BESS which make these technologies increasingly agile in combination with increasing the integration of wind and PV generators. BESS is playing a significant role in the transition from traditional synchronous generation to 100% renewable generation [63].

They are considered complementary energy storage in high share VRE systems due to the aggressive cost reduction of this technology [51]. For example, a considerable cost reduction was recorded in Li-Ion batteries over the last 10 years (from about 1000 $/kWh in 2010 to <200 $/kWh in 2020) [46]. In principle, batteries belong to electrochemical technology. In Ref. [66] several battery technologies are discussed from lead acid to Li-Ion batteries. However, Li-Ion batteries are the most popular and promising storage technologies in the market for

Table 2
Techno-economic parameters of key storage technologies.

Storage type	Capacity (MW)	Energy density (Wh/L)	Power density (W/L)	Efficiency (%)	Response time	Lifetime (Year)	Capital Cost ($/kWh)
PHES	10–5000	0.5–2	0.5–1.5	70–85	s ~ min	40–60	5–100
CAES	3–1000	3–6	0.5–2	70–90	≤15 min	20–40	2–200
FES	0.1–0.25	20–80	1000–5000	70–93	≤4 ms	15–20	900–12000
HES	0.1–50	500–3000	>500	20–50	<1 s	5–15	10–20
Li-ion	1–100	95–500	56–800	85–98	ms ~ s	5–16	120–1000
Lead acid	0–40	150–345	140–180	70–90	ms ~ s	5–15	270–400
Flow cell	0.1–50	>500	>500	60–85	ms ~ s	5–20	350–2000
SMES	0.1–10	0.2–2.5	1000–4000	95–98	≤10 ms	>20	10000
Supercapacitor	0.3–50	10–30	100000	90–95	≤10 ms	>20	20000
TES	0–300	80–500	–	30–60	≤10 ms	5–40	3–5

*Note: Energy density and power density can be also represented in J/m^3, J/L.
*1Watt (W) = 1 J, 1 Wh = 3600 J, and 1 m^3 = 9 1000 L, where L is Litter.

Table 3
Summarises the global BESS projects which are in operation on the islands power systems.

Country	Project name	Power/energy	Application	Commission date	Ref.
Ireland	RWE BESS project	60 MW/60 MWh	Facilitate the integration of wind power, supporting grid stability and reliability	2022	[78]
	Lumcloon BESS project	100 MW/100 MWh	Providing reserve and frequency regulations	2021	[78]
Texas	Rabbit Hill	10 MW/5 MWh	To provide ancillary services and energy optimization	2020	[79]
	Odessa, Snyder and Sweetwater BESS projects	3 × 9.9 MW/20 MWh	to address grid stability issues in rich wind area, manage grid operation	2021	[80]
Australia	ESCRI South Australia Project, Dalrymple	30 MW/8 MWh	Ancillary service and energy market	2019	[81]
	Hornsdale Power Reserve, Jamestown	150 MW/194 MWh	Ancillary service and energy market	2020	[81]
Japan	Hitachi Energy Faroe Islands BESS project	6.25 MW/7.5 MWh	Power backup, increase the utilisation of the wind farm	2021	[82]
Northern Ireland	AES Kilroot	10 MW/5 MWh	Frequency response and operating reserve services	2021	[79]
Portugal	EEM's BESS project on island of Madeira	15 MW/16.4 MWh	Increasing renewable generation mix, Black start, system restoration	2022	[83]
	A pack of lithium-ion batteries Island of Porto Santo	4 MW/3 MWh	Maximize renewable integration, frequency regulation, improve power system performance	2019	[84]
Hawaii islands	BESS project in Kauai Island	13 MW/52 MWh	To replace some diesel peaking plants during evening and run the Kauai grid on 100% solar during sunny day.	2017	[85]
United Kingdom	Hertfordshire BESS project	50 MW	Grid ancillary services	2018	[86]
	Pillswood project	98 MW/196 MWh	To provide critical balancing services and maximizing the integration of wind powers	2023	[87]

both portable devices and energy storage [67]. This is because of their high efficiency, high energy density; low self-discharge; lightweight, flexibility, and relatively low maintenance in comparison to other battery technologies.

Although the BESS has no geographical restriction, there are some weather-related limitations. For example, manufactures recommend maintaining the operation temperature of Li-ion battery between 25 °C to 30 °C [68]. Any deviation in this range may adversely affect on the performance and lifespan of the battery. However, this challenge is not considered as a geographical limitation for exploring the superior application of BESS technologies. Currently, advances in energy management systems and cooling technologies (i.e., ventilation and HVAC) are widely deployed to mitigate this challenge. Therefore, it can be installed at all locations within the power system, from grid level by TSO to behind metering by residential users.

The installation of battery storage systems is increasing at an incredible rate across the globe due to the national ambition of many countries towards 100% renewable energy and international obligation to meet Net Zero GHG emissions by 2050. According to the data available in Ref. [69], the total BESS capacity installed (at the utility scale and behind the meter) was 17 GW by the end of 2020. It is expected that total battery storage capacity will reach about 600 GW in 2030 to meet Net Zero GHG emissions targets [69]. According to Ref. [51], Australia is a global leader in terms of BESS power capacity with 246 MW deployment (at single sites 100 MW/129 MWh batteries installed in Southern Australia), whereas the USA is a leading country in terms of BESS energy capacity with 431 MWh capacity.

There have been numerous studies of BESS providing grid services including ancillary services, voltage regulation, frequency support, and time shifting [6,23,48]. They are in huge demand for fast frequency response (FFR) applications due to their fast power delivery to the system within a few hundred milliseconds following a disturbance, as required in Ireland [33] and the UK [70]. The UK national grid plans to install 362MW/391 MWh BESS for FFR services [71]. With reducing frequency response time, BESS could potentially be a solution for RoCoF or inertia services [72], and mimic synchronous inertia service [33]. These types of services are crucial to facilitating a high share of VRE generation and preventing undesirable events such as the blackout in South Australia in 2016 [70].

According to a study conducted by Ireland's TSO [73], if appropriately controlled, 500 MW of BESS would facilitate the secure operation of the Irish power system at 75% renewable generation. The impact of BESS on RoCoF and frequency is investigated under actual generator losses in the Irish power system in Ref. [72]. It concluded that 3 GW of synchronous generators can be replaced with only 360 MW of BESS if the response time delay of BESS is appropriately controlled to be less than 100 ms after a disturbance is initiated and to reach full active power response after 450 ms.

Battery deployment globally has grown to match VRE penetration. For example, in the Republic of Korea, several aggregated BESS ranging from 16 to 48 MW are installed for frequency regulation services [74], in Japan, there is a project to install 240 MW/720 MWh BESS combined with a 500 MW wind farm to provide generation-demand balancing and grid stability services [75] and in China, there is a contract in place to install 30 GW BESS to facilitate more VRE generation integration by 2025 [76]. BESS will play a significant role in maintaining the frequency stability of the power system with 100% intermittent renewable generation [17].

The utilisation of BESS is widely used in island energy systems to enable the transition to 100% VRE system, while supporting the whole system power balancing and providing system ancillary services requirements [77]. In such systems, the conventional power plants are gradually replacing with wind and solar power generation. Table 4 presents the summarised BESS projects in some island grids around the world. Table 5 summarises some current synchronous condenser projects in some islanded grids.

BESS can mimic the droop response of conventional synchronous generators. For example, the efficient response of 100 MW/130 MWh BESS (based Lithium-ion battery) to two extreme frequency events in South Australia is globally well known for providing 70% of its capacity reserve for FFR support in low inertia conditions, while using the remaining 30% of its capacity for conventional energy market services [43]. Fig. 3 presents the actual BESS response for both events: (a) BESS response for low frequency event on 25th Aug 2018, (b) BESS response for high frequency event on 31st Jan 2020.

It can be clearly seen from Fig. 3(a) that BESS was able to inject active power rapidly in response to a local frequency reduction to 49.85 Hz, and subsequently started to absorb local system active power after local frequency recovery exceeded 50.15 Hz, and finally BESS reverted to absorbing active power output when the local frequency recovered to

Table 4
Worldwide established EV project capabilities for system ancillary services provision.

Country	Frequency regulation	Reserve	Distribution services	Arbitrage	Time shifting	Emergency back-up
USA	x	x	✓	X	✓	x
China	✓	x	x	X	✓	✓
Germany	✓	x	✓	✓	✓	x
UK	✓	✓	✓	✓	✓	✓
France	✓	✓	✓	✓	✓	✓
Italy	✓	x	x	X	✓	x
Portugal	✓	x	✓	✓	✓	x
Switzerland	✓	✓	✓	X	✓	x
Netherland	✓	✓	x	X	✓	✓
Belgium	✓	✓	x	X	✓	✓

Table 5
Current synchronous condenser projects in some islanded grids.

Country	Project name	capacity	Objective	commission date
UK	Phoenix Project in Neilston by ABB project partner	70 MVA in combination with 70 MVA STATCOM converter	To provide reactive power support for black start	2022
	Rassau Project in Wales by Siemens Energy and Quinbrook	100 MVA	To ensure power system stability while approaching net zero target	2024
Denmark	Suðuroy, Faroe Islands Project by ABB	8 MVA in combination with 7.7 MW BESS	To enable 100% VRE generation in Suðuroy power system	2022
	Streymoy, Faroe Islands by ABB	8 MVA	To enable 100% VRE generation in Streymoy power system	2023
Australia	Energy Connect, interconnector by ANDRITZ	Four SCs plants with capacity of 120 MVA/ plant	To support system inertia	2023
Ireland	Green Atlantic at Moneypoint by Siemens and Electricity Supply Board (ESB)	400 MVA	To accelerate the higher share of VRE generation	2022

its nominal operating frequency limit. The same scenario happened in Fig. 3(b) with the BESS rapidly absorbing reactive power in response to the increase in local frequency.

3.2. Demand response

Power mismatch between supply and demand results in frequency instability, inadequate generation reserves, and insufficient equipment responses [88]. The increase in VRE has seen increases in the frequency of occurrence of mismatches. In traditional power systems, spinning reserve from fossil fuel generators is increased/decreased to deal with the imbalance between supply and demand, which increases costs and GHG emissions.

In Milo et al. [17] the utilisation of residential and commercial loads to provide a system reserve for frequency regulation is investigated. Demand side participation has been used in different forms for decades to lower, increase, or shift load demand in power systems with generation constraints [89], such as peak shaving and valley filling [90], and ancillary system services [91]. These techniques have been used in ERCOT to manage system stability with a high share of VRE systems [26].

In a demand response (DR) program, a group of loads is turned on/off to provide reserve capacity and flexibility [92]. Several electronic devices and communication technologies have been utilized in modern power systems to enable a high level of flexibility and controllability on the demand side. This has facilitated the participation of certain types of loads to meet the grid's requirements [88]. In Ref. [93], DR was used as a smart appliance for frequency-controlled reserve using programmable thermostats. The experiments conducted on thermostatically controlled 10 kW loads showed that there was a significant relationship between the power consumption of each load and frequency response. This is because loads were acting as a reserve for regulation up and down. Detailed discussions of DR capability in system frequency regulation and associated computer programming control algorithms can be found in

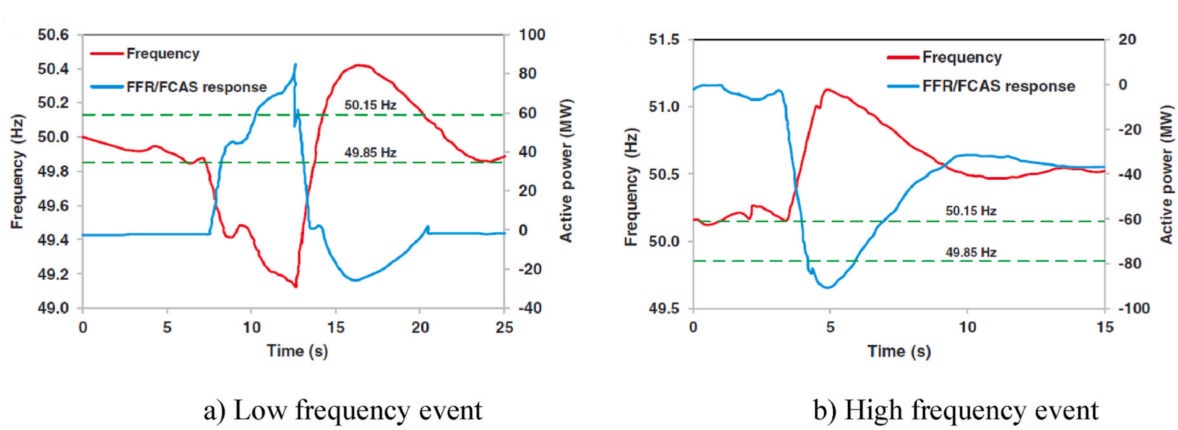

Fig. 3. South Australia's actual BESS response for low and high frequency events [43].

Refs. [88,92].

In [25], demand response programming is used to observe the influence of customers on system voltage stability and concluded that voltage stability improved with increased participation of customers. One way to improve steady state voltage management is improving voltage drop at the end users buses by reducing demands on related feeders [23]. In addition, inductive and power electronic controlled loads can be managed to enhance reactive power consumption and improve grid voltage stability.

The important contribution of smart appliance technologies is expected to increase with the increasing penetration of VRE in the future since flexibility in controllable loads can effectively contribute to power balancing management between generation and demand, and hence power system voltage and frequency regulation [93]. Another application of smart appliances is to shift system power demand to coincident periods with intermittent generation output of VRE [25]. This application will lower the ramping requirements from grid utility and reduce wind and PV power curtailment.

Residential grid scale energy storage, industrial load (including data centres) electric vehicle charging, and heating/cooling systems which are inherently flexible loads can facilitate cost-effective shifting of peak demand [23]. For example, data centre load demand is anticipated to be 30% of Ireland's total electricity demand by 2030 [94]. This is really a huge potential load to be controlled for supporting system voltage and frequency regulation.

Integration of other smart appliances (i.e., smart charging of EV, thermal loads, boilers) can offer additional flexibility to achieve a high level of system services (i.e., ancillary service and reserve) from the demand side [95]. In the UK, heating which mainly relies on gas boilers accounts for 40% of total energy consumption by twenty-four million end users [96]. Electrifying this sector using a heat pump can increase system security as well as reduce GHG emissions.

For example, centralised large scale heat pumps or aggregated distributed small scale heat pumps can be directly controlled to provide ancillary service (i.e., FFT) and reserve service (i.e., demand turn up) [96]. Using a decentralised control algorithm, Muhssin et at [97]. investigated the potential of aggregated controllable loads including domestic heat pumps and fridges for frequency service provision in the UK power system. The study concluded that aggregated heat pumps and fridges can reduce the need for generators spinning reserve in the UK power system by 50%.

In Germany, there will be a 150 TWh demand (around 25% of the total electricity demand) for electricity from transportation sectors if they are totally electrified [98]. With proper control, EVs can be aggregated in the form of a virtual power plant (VPP) to contribute to power balancing, voltage frequency regulation, backup power, reserve capacity, and VRE curtailment reduction [99].

3.3. Electric vehicle

Electric vehicles (EV) can be used to provide power system ancillary services support when they are linked to the grid through home based-charging units or public charging stations using vehicle to grid (V2G) and or grid to vehicle (G2V) controls [43]. The global sales of EVs exceeded 6.4 million in 2021, with a growth of more than 98% in comparison to 2020. The figure was 4 million units for battery electric vehicles (BEVs) and 2.6 million units for plugin hybrid electric vehicles (PHEVs). The EU (including the UK) is expected to hit 14 and 33 million EVs by the end of 2025 and 2030, respectively [100]. These 'mobile' battery storage devices are commonly known as V2G to G2V technology.

The dynamic operation of grid stability may be negatively affected by stochastic charging and discharging of EV batteries. For example, an unprecedented increase in charging infrastructure will result in unexpected peaks in demand, voltage sag, imbalance phase, and system collapse [23]. However, EVs can be well programmed and controlled to provide grid stability services including voltage regulation and frequency response services during normal operation and after abnormal events once the correct controls and grid infrastructure are in place [101].

To increase the ability of EV charging to improve grid stability, advanced control mechanisms (i.e., bidirectional battery charging units) and optimization algorithms to monitor active and reactive power dispatching, and frequency and voltage control, once they run over a predefined threshold limit, are needed to facilitate V2G and G2V services [43]. Considering system stability problems, the location and size of EV charging stations or points must also be determined [23], so that V2G and G2V coupling can offer ancillary services to enhance grid stability. In Ref. [102], a dynamic vehicle grid support control algorithm was used to control EV charging stations in response to frequency events. It was observed that aggregated EVs reduced frequency deviation from 49.1 Hz to 49.57 Hz in 8 s.

To obtain grid ancillary services from V2G or G2V deployment, the EV batteries' state of charge (SoC), rate of charging (A/s), incentive cost, and voltage and frequency threshold (to trigger these services) must be considered. A stochastic dynamic programming framework was used to study home energy management with EV battery storage [103]. Results show that with proper charging coordination, plug-in EVs can improve voltage stability, system reliability, and energy costs by decreasing power losses and improving the distribution system load factor. There are many global established EV projects which can offer different system services [161]. The current capabilities of worldwide EV pilot projects are summarised in Table 3.

3.4. Synchronous condenser

A synchronous condenser (SC) is a synchronous motor running under no mechanical load. Although it is not new technology the role of a SC is redefined to tackle challenges associated with the integration of non-synchronous VRE generation in terms of system inertia reduction and low short circuit current strength. By controlling the field winding excitation, SCs are capable of injecting/absorbing reactive power, improving the power factor, and supporting system voltage control [104].

According to Ref. [43], modern SCs, in co-existence with flywheel capacitors, are able to release stored rotational energy for a short time, and they can also support short circuit current as well as deliver reactive power requirements. In Ref. [105], SCs are proposed for improving grid voltage stability in low short circuit strength network regions. Reduced system inertia and frequency stability concerns due to the installation of non-synchronous VRE generation can be mitigated using SCs [5]. SCs are suggested as a complementary response service to solar PV, wind, and battery energy storage plant response while providing reactive power [26].

The role of SCs in the Western interconnection in the USA is investigated in Ref. [7]. Results show that increasing the share of wind and solar up to 66% of total load demand and reducing 90% of coal power plants resulted in power system separation. The system was able to return to acceptable dynamic operation performance by replacing only three offline (1.4932 GW) coal power plants with online SCs [106]. This capacity is equivalent to less than 5% of the total system generation.

Locating SCs close to solar and wind power plant generation is important to grid stability. Several SCs are currently installed in Germany, the USA, and South Australia to improve system strength [105]. Due to the significant share of VRE, the Danish power system is another example of widespread utilisation of SCs to support system rotational inertia and short circuit currents during fault events [34].

However, in contrast to their ability to provide multiple grid stability supports, SCs are more likely to create sub-synchronous oscillations, due to their control interactions with local solar-PV and wind power plants [107]. This is mainly due to the interaction between mechanical shafts and electronic devices. Therefore, it is extremely important to perform electromagnetic transient studies of the system under different operating conditions before installing this technology in weak power networks [43].

The synchronous condenser has several considerable technical capabilities over IBR resources such as the ability to maintain terminal voltage, high short circuit injection and inherent inertia response [35]. These technical capabilities cannot be replaced by power converters. Therefore, the installation of SCs is crucial to ensure the security and strength of islanded power systems. Current SCs projects are summarised in Table 4.

3.5. High voltage direct current (HVDC) interconnectors

HVDC interconnectors are used to transfer power over a long distance at low power loses without technical limitations. HVDC transmission systems are mainly used to connect offshore wind to the grid and cross boarder interconnections. These transmission systems are classified as asynchronous and embedded interconnectors [108]. The former is used to connect two AC lines that have different frequencies or phase angles such as the HVDC line between Norway and Netherlands. The latter is used to connect to synchronous areas such as the interconnection between Denmark and Germany.

Currently, more than 10% of the EU's generation capacity is from interconnectors, the European Commission set a plan to increase interconnector capacity to 15% by 2030 [109]. With correct regulations to support cross-border energy trading, this will facilitate the share and trade of more electricity among European countries. Interconnectors allow each EU member to have access to a broad range of electricity generation sources (i.e., wind, solar, and hydroelectric power) from other EU members. This will increase the reliability of power supply and decrease the need for fossil fuelled power plants [110].

In addition, interconnectors can facilitate the share of surplus renewable generation between countries. For example, demand shortage in one country, due to low sunshine or wind speeds, can be filled by importing surplus electricity from neighbouring countries where generation is higher than demand. The electricity trading over interconnectors can add extra generation capacity to participate in the balancing market in each country in the EU.

Recently, Danish Government promoted the world first large-scale energy islands located on an artificial island in the North Sea and at Bornholm, where they are expected to generate 12–13 GW of offshore wind power by 2040 [111]. The main objective behind the projects is to serve as green power plants to generate power from surrounding offshore wind farms and distribute it to Danish power system and its neighbouring countries through HVDC interconnectors [112]. This is not just increasing the security of supply of Danish electricity grid, but it will also increase the generation capacity European countries as they have strong interconnectors with Denmark. The islands can be further expanded to generate more green power and transfer it to anywhere in the world via interconnectors. To facilitate further expansion of these projects, the Danish government will add Power-to-X technologies and batteries to the energy islands [111].

HVDC can provide a broad range of ancillary services including frequency response and reserve, and reactive power reserve. According to the study in Ref. [113], by controlling the power flow between two synchronous areas, HVDC can support frequency regulation if a correct fast frequency control is utilized. The role of interconnectors is crucial to maintaining power balance during high VRE generation. For example, wind generation exceeded demand by 40% in Denmark on a summer day in 2015, and interconnectors were allowed to share 100% of surplus power with Germany, Norway, and Sweden [108]. Without having an interconnector with these countries, Denmark's grid might face over frequency followed by cascaded trips and blackouts.

In low inertia systems, FFR is necessary to respond to changes in frequency before the primary frequency control, unless it takes a shorter time for system frequency to reach its critical value after faults or trips. To prevent faster frequency droop to below threshold value, HVDC interconnectors are widely used by most of the power systems to connect with other systems. HVDC interconnectors are able to quickly change their output real power and provide FFR services at low cost [114]. As a complement to conventional frequency control, the potential economic benefits of utilising HVDC interconnectors for fast reserves provision are studied in Ref. [115]. The study concluded that using HVDC interconnectors for FFR services can reduce the cost of security for Nordic TSO by 70%.

Interconnectors facilitate bidirectional power flow over wider geographical areas. This is crucial for enhancing system adequacy, flexibility and security of supply, power balance, and installation of more VRE generation [116]. However, rapidly changes in the power flow direction, which depends on a marked price signal, can also create negative impacts on the system's security without introducing appropriate regulations. An instantaneous deficit between supply and demand was recorded in the UK due to an abrupt change in the power flow direction in response to the price signal [116]. Therefore, this has to be mitigated by introducing proper grid codes and standards.

HVDC interconnectors can inject/absorb reactive power and support voltage regulation if utilising the right control technologies. For instance, HVDC based voltage source converters (VSC) can be designed with inherent reactive power compensation to support system voltage without the need for additional equipment [109]. HVDC based VSC can instantaneously respond to network disturbances and thus improve voltage stability by maintaining system voltages within operational limits during large trips and faults [109]. Results of [117] showed that HVDC based VSC can restore an AC system from a total shutdown state after a large disturbance and provide reactive power similar to STATCOM.

If it is built with the correct grid forming converter technologies, HVDC can provide faster reactive power control. The authors [118] evaluated the performance of HVDC based VSC for transient events. When grid following converter control was implemented, voltage oscillations were present in the system. It was found that the system voltage dynamics and transient time can be improved significantly if GFM control is implemented instead of GFL.

Inverter based resources (IBR) are also interfaced with the grid through GFM and GFL converters. Therefore, building DC distribution system can reduce the losses and cost, as the VRE can directly connected to the DC network without further conversion stages [119]. In comparison to conventional AC network, DC systems can transmit higher power over the existing AC line, improve power quality, and increase system safety.

For example, medium voltage DC (MVDC) network can be used as a collection grid for offshore wind farms or solar power plants over wide area and connect them to a low-level DC network [120]. MVDC system is acting as the same way as HVDC system, but on a smaller scale and over shorter distances. Therefore, building DC networks can directly facilitate the integration of VRE generation. Despite these advantages, there are serious concerns about the application of DC grids such as their protection, fault current behaviour, fault location, and voltage stability [119]. Although, they are considered as an alternative to AC grids, the employment of DC networks needs to be studied further which is out of scope of this study.

3.6. Enhanced inverter control technology

The integration of non-synchronous VRE generation in today's power system is increasing as power systems move towards a Net Zero GHG emissions target by 2050. The extensive introduction of power electronic grid interfaces will bring power system stability challenges in terms of reduced damping to the grid as conventional fossil fuel power synchronous generation retires. Power systems with such generation portfolios will experience low frequency nadirs, high RoCoF and decreased power system stability [7,41].

Conventional commercial converters were designed to incorporate a current control that will not allow them to participate in most of the regulation services [119]. To get ahead of this reduced performance, moving from a grid following inverter to a restructured grid-forming converter control strategy is suggested, as this can mimic the behaviour of synchronous machines to support grid stability [5,120]. Grid forming control technology allows IBR generators to support frequency and voltage and improve angle stability in the power system [119].

Grid-following (GFL) inverters are widely used with most of today's VRE generation installations. A grid-following inverter acts as a current source to inject real and reactive power into the main power grid. The philosophy is that a phase locked loop is used to measure the angle of the grid voltage as an input signal to the GFL so that the connected generators can follow the grid voltage [7]. Grid following cannot directly support system frequency and voltage regulation because they rely on something else to form the system voltage as they follow the existing grid voltage at the point of common coupling (PCC). This is a very effective control strategy to connect VRE generation in a system dominated by conventional fossil fuel generators as the high rotating mass inertia of conventional power plant enable them to form a suitable sinusoidal waveform and respond to the system stability challenges [120].

However, with the increasing penetration of variable renewable generators, system inertia is reducing dramatically as conventional power plants are replaced with inverter-based resources. Therefore, the transition to 100% inverter-based renewable power systems requires new technology to offer the same services provided by traditional synchronous generators particularly inertia response and short circuit current requirements.

In addition, it is a fundamental requirement to install new technologies that produce the grid frequency and voltage to secure stable operation in a VRE dominated power system. In the literature, a variety of grid forming (GFM) control strategies are proposed to tackle weak grid stability issues such as droop control, virtual synchronous machine, and virtual synchronous oscillators [26,121].

Grid forming converters act as a voltage source to set grid frequency and voltage. The behaviour of GFM converters is similar to synchronous generators and enables independent system operation because they create their own internal reference voltage angle based on the converter's output power [122]. Potentially, GFM converters can provide faster grid stability services because their response to abnormal conditions comes naturally and is not subject to delay [26].

It was observed that there is no significant delay between the change in output power and frequency response of droop controlled GFM converters while responding to any contingency [122]. This is because GFM inverters do not rely on an external parameter to form them as they create their reference voltage. The main difference between both GFL and GFM control structure is illustrated in Fig. 4.

In the last decade, GFM inverters have been introduced in power systems to tackle the challenges associated with inertia deficiency in order to provide damping to changes in system frequency [5,17]. Recently, the capability of GFM inverters has been tested in some islanded power systems without using conventional synchronous machines such as the island of Tokelau (in 2013) and Ta'u (in 2016) [26].

In [123] a GFM based solar PV system with a voltage source converter is shown to be able to emulate the behaviour of synchronous generators to provide active and reactive power and support fast frequency response to the grid. The results in Ref. [120] indicate that GFM converters show similar behaviour to synchronous generators and are able to provide frequency support services in the form of synthetic inertia to the grid. With the aid of these technologies, wind, solar PV, and energy storage can provide all stability services provided by a conventional power plant and even offer superior fault ride through performance.

Although GFL technology is able to offer some important stability services such as FFR it may suffer from PLL instability-related issues under weak grid systems. The literature indicates that GFM converters can offer more system robustness and stability services under low inertia conditions [124]. Chen et al. [125] suggest that in the worst scenario 30% of VRE must be connected using GFM interfaces to ensure system reliability and stability.

However, the percentage of GFM based resource share in the VRE dominant power system needs to be studied at a power system level. Deployment of a large amount of GFM based resources in modern power systems may be a challenge due to the potential control interactions between individual GFM inverters nearby [126]. Grid forming inverters will create a complex dynamic control interaction with nearby synchronous generators due to their fast response time and interdependent control nature [43].

Recently, such control interaction issues were reported in the USA, Texas, and China [127]. Control interactions, which result in power and voltage oscillations, are likely to occur in the rich VRE grids. Therefore, GFM technology needs substantial focus to future-proof 100% VRE power systems. Although GFM converters are capable of providing all stability services provided by synchronous machines, GFMs have not been implemented by power system operators at a large grid scale under 100% VRE instantaneous operation.

The GFM technology needs to be further studied to analyse, solve and address some of the limitations of GFM technology with regard to handling high short circuit current for a short duration and energy buffer availability (i.e., the storage state of charge, available MWh) [43]. Dynamic simulation study analysis to investigate the adverse control interactions between individual GFM inverters with other GFM and GFL inverters-based devices is crucial for future grids [126]. These challenges must be managed in grid code specifications.

Both GFL and GFM are subjected to physical constraints (i.e., energy, current, and voltage limits), mechanical constraints (i.e., wind turbine generators), and external constraints. These constraints need to be managed through grid codes. Currently, only a few manufacturers supply commercial-grade GFM. This is because manufacturers do not have clear specifications or demand for developing GFM. There is no international or national standard on GFM functionality. The European

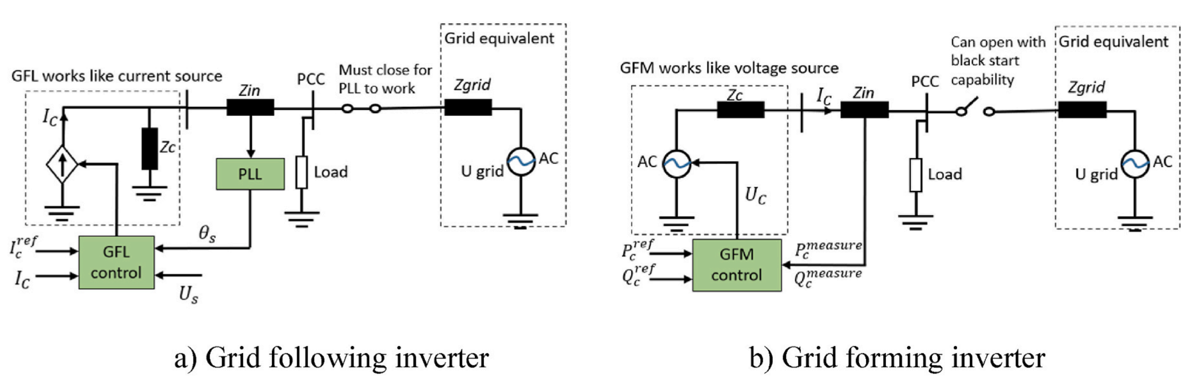

Fig. 4. General control structure of grid following and grid forming inverters.

Network of Transmission System Operators for Electricity (ENTSO-E) defined several technical standards for GFM capabilities to meet future power needs [128], as presented in Fig. 5.

To the best of the authors' knowledge, the UK is the only country that has published minimum grid code specifications for grid forming capabilities to meet future power needs [119,129]. These specifications have been defined in GC0137- published by National grid ESO [130]. Grid codes and technical standards for GFM are not defined by almost all other countries. For example, there is no technical standard for GFM converters in the latest edition of the Italian CEI 0-16 and 0-21 standards, published in 2022 [131], the EirGrid Grid Code Version 10, published in December 2021 [132], and the Spanish Technical Supervision Standard, 'Norma Técnica de Supervisión' (NTS), published at the end of 2020 [133]. Although GFM capabilities are not defined in grid codes, several GFM projects are in operation in the world for providing different system services [134].

3.7. The role of electricity market

The power balance between generation and consumption must always be maintained to ensure the reliability and stability of power system operation. In this context, system operators introduce different ancillary services (i.e., voltage and frequency regulations) as well as capacity reserve markets. Currently, most of these services are inherently provided by fossil fuelled synchronous generators to maintain system reliability [135]. However, power systems transitioning to 100% renewable electricity means that the traditional system services must be provided by other means.

The power generation transition to new forms of electricity causes fundamental changes in how the power system is managed. These trends must also be reflected in the electricity markets. Future electricity markets should be designed and operated to support a higher share of VRE, smart energy systems, and sector coupling to accelerate the transition to 100% renewable electricity [136]. This requires liberalizing the energy sector to enable DERs to participate in ancillary services markets to expand revenue sources and assist system operators in scheduling an optimal generation mix. Existing electricity markets must also be adjusted to increase the impact of demand participation, and scarcity pricing, and to expand the wholesale electricity market competitive model at the distribution level [137]. Flexible generation can provide instance balance during load changes and reduce the potential causes of system failure.

Technologies (i.e., EVs, heat pumps, storage) are effectively becoming a component of the grid when they are connected to the power system. Thus, they can provide more flexibility to the grid by adjusting their electricity demand profile according to the wholesale market price signals and decoupling the timing of energy demand for these technologies from electricity demand [138]. Flexibility in modern power systems is a key driver to facilitate the integration of high levels of inverter-based resources while ensuring the system's reliability and stability at a reasonable cost [139].

Maximum flexibility of the power system must be obtained based on current and ongoing innovations in market design, business model, and system operation's enabling technologies [140]. With the incredible deployment of VRE, unit commitment (UC) is actively studied by researchers and operational engineers to identify the least cost generation mix to meet system security constraints (i.e., voltage and frequency stability) throughout system flexibility enhancement [141]. However, enhancing system flexibility is combined with the diversification of energy resources while moving to 100% VRE systems. Current market structures were not designed to deal with variable generation mix [138].

Unit commitment is widely deployed to ensure resource adequacy to meet demand and to maintain system reliability in the presence of unexpected changes in operating conditions [142]. In several studies, UC models are developed to address the negative impacts associated with a high share of VRE on power system stability and reliability [142–144]. For example, inertia and primary frequency control constraints are included in the unit commitment model developed in Ref. [143]. The proposed model was used to increase the spinning reserve requirement by arresting frequency nadir to be above a threshold limit for every hour of the UC horizon. Results show that the UC solution is able to facilitate the integration of high levels of VRE while avoiding load shedding after contingency.

Considering the participation of controllable loads, the UC model is proposed for scheduling demand resources in response to transmission congestion events [145]. Study [142] presents a suitable methodology to capture variable behaviour of wind generation on transmission lines using UC in combination with power flow analysis and historical time-series data. Results show an improvement in steady state voltage stability of the Irish power system under a high share of penetration VRE. In Ref. [144], a Monte Carlo simulation in combination with the unit commitment problem was developed for voltage control in transmission grids by enabling optimal reactive power flow/dispatch decisions in response to changes in bus voltages.

In the future net zero emissions markets, DERs can be intelligently managed to support system ancillary services through price-based incentives [146]. The concept of a virtual power plant (VPP) is widely regarded due to its capability to aggregate controllable DERs (i.e., storage, EV, and Heat pumps) and its potential to increase grid flexibility [147]. Using embedded technologies and communication networks through bidirectional power flow, VPP enables end users to trade excess electricity on the market at a reasonable price without the participation of a third party [147].

The UK, USA, EU, and Australia are the global leaders in the VPP market [148]. The worldwide VPP market is anticipated to reach $5.9 billion in 2027. Virtual power plants are able to operate in several electricity markets, ancillary services, trading energy and other financial products similar to fossil fuel-based power plants [149]. For example, as virtual energy storage, EVs have been aggregated to support voltage and frequency control in the electricity market based on incentive prices [150].

Therefore, rethinking electricity markets is required to facilitate bidirectional energy flow marketing among prosumers, suppliers, and consumers. Information and communication technology (ICT) will be

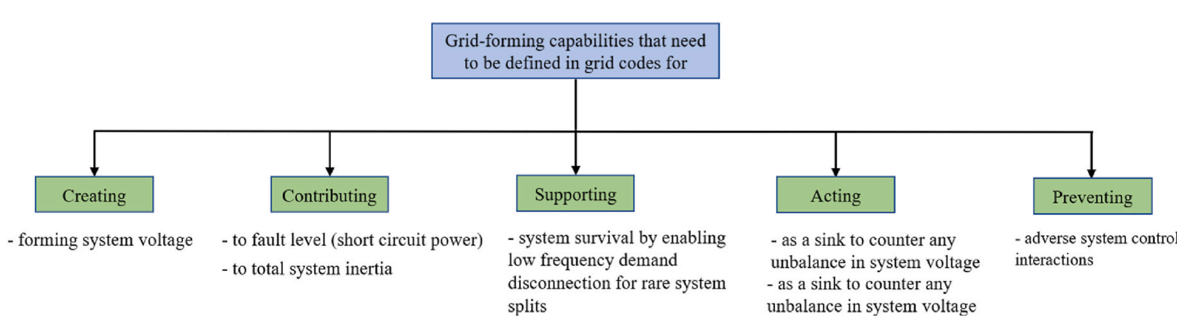

Fig. 5. GFM capabilities to meet future system needs.

the heart of future electricity markets as a key enabler for facilitating bidirectional power flow based on control signals and price incentives, as well as consumer participation in the wholesale markets.

3.8. Cyber security

The transition from centralised, unidirectional power flow with few players to decentralised, bidirectional power flow with multiple intelligent players on both the generation and consumption sides will involve a significant amount of data and control signals. Controlling bidirectional power flow and maintaining power balancing based on market price signals requires advanced digital technology, ICT, and management infrastructure [151]. This smart infrastructure will coordinate message transfer between dispatchable components, controllers, and sensors using communication means (i.e., routers, switches, and wireless links). Given the physical application and infrastructure domains, the security of such a smart system with a massive data flow is a major concern, due to the vulnerability to cyber-attacks [152].

For example, denying actual commands or injecting fake commands from communication links can deceive the control centre and cause the system objective to become disoriented [153]. The timeliness of key orders is critical, as delays in providing signals responsible for triggering protection devices or asking for frequency response are more likely to cause instability or even a blackout [151]. Historically, the first cyber-attack on Ukraine's power system was recorded in 2015, resulting in a power outage for 225,000 consumers for many hours [154].

Cyber security can play a vital role in ensuring the security and stability of smart grids [155]. It is a set of tools and techniques to protect computerised elements against damage and incursion. This is projected to be a fundamental part of a 100% renewable electricity system, ensuring the resilience and reliability of the system. A number of international organisations, including the USA national institute of standards and technology (NIST), the International Organization for Standardization (ISO), and other standards are collaborating to better understand emerging cyber-attack threats and establish a set of cyber security standards to confront cyber-attacks in modern power systems [154].

As a most successful way for improving cyber security, resilience, and reliability of smart grids, machine learning based-real time intrusion, network segmentation, and artificial intelligence are widely used [152,153]. In Ref. [156] a cyber security plan was designed using machine learning techniques in conjunction with the Internet of Things (IoT) to allow risk assessment and real-time decision making in smart grids. The developed models were successful at overcoming cyber-attacks.

4. The role of grid codes and interconnection standards

A national grid code is a key mechanism used to manage power systems globally to ensure safe and reliable power system operation and interconnection procedures when connecting new technologies [157]. Recent development in inverter-based technologies brings new challenges to the design and implementation of system operation grid codes. The main challenge in modern power systems is the ability of VRE to ride through voltage dip and frequency deviation.

Grid code requirements vary from one country to another because they are determined based on the specific national system generation characteristics and transmission operating requirements [23]. National grid code specifications are currently subject to continuous updates to deal with power system changes including immediate action requirements on the transition pathways to 100% VRE generation.

Modern power systems are evolving in several ways as new inverter interface technologies (e.g., wind, solar PV, solar rooftop, batteries, and smart appliances) increase in penetration. These technologies are expected to be smart and able to communicate with each other and the grid fully. The grid needs more flexibility to handle all these technologies, which can be obtained by enabling/leveraging these smart technologies in the right way and by advancing grid codes. IEEE standard-1547, which is the fundamental international document for integrating DER technologies, has been updated several times during the last 30 years [158].

A number of international grid code requirements for DER technologies with fault ride through (FRT) capabilities have been published such as European grid code EN 50438:2013 and EN 50549:2019, Italian grid code CEI 0-21, German grid code VDE-AR-N 4105: 2018-11, Great Britain grid code EREC G99-2018, and Canadian interconnection standard CSA C22.3 No. 9-2020 [157]. Although these standards and grid codes have shown a positive impact on grid stability, they are not defined for a system with 100% variable inverter-based generation.

IEEE smart grid is a unique technical standard guide for interoperability[1] standardisation of smart grids which is updated regularly by a panel of experts from IEEE technical committees [159]. These committees develop and publish a series of standards as a foundation for connecting and operating diverse technologies in modern power systems including the interconnection of renewable enabling technologies (i.e., wind, solar, energy storage, smart appliances), control technology, reliability, security, digital information, sensors, and technical assessment [157].

Currently, IEEE has 100 published, and in development, standards to address the safety and interoperability of smart grids. Among these 20 standards are named in the NIST framework standards for smart grid interoperability [159,160]. Some other organisations are publishing a series of international standards for electrical power systems including smart grid interoperability such as NIST [160], the international electrotechnical commission (ICE) [161], and ISO [162]. These standards provide best practice and concrete solutions for adopting renewable enabling technologies by ensuring the interoperability of smart energy systems while transitioning to 100% renewable electricity systems [161,162]. Following these standards facilitates the communication between various components of a modern power system, hence increasing the system's security and reliability.

These standards are updated continuously to increase the ability of inverter-based resources to provide a variety of power system stability supports, thus facilitating the transition to 100% renewable electricity systems [157]. For example, IEEE Std 1547 has been updated several times to increase the performance and functional capability requirements for grid connected DERs since 2018. The last version of IEEE 1547 (P1547.9) mandates the DER to provide power system support, a guide for the connection of EV charging stations, allow bidirectional active and reactive power flow, and interoperability of DERs with associated power system interfaces [163].

In addition, the ISO 50001:2011 which was an energy management standard, has been updated several times [164]. ISO 50001:2018 is the latest version of this standard which provides best practice in energy efficiency through systematic energy management to improve performance and reduce consumption [165]. Furthermore, IEC TR 61850-90-1:2010 was first published to define the communication protocol for smart electronic devices in power utility automation [166].

The latest version of IEC TR 61850-10-3:2022 is considered a core standard for intelligence and autonomous grids [161]. It includes the communication structure for digital technologies and the control architecture for the connection of DERs and interoperability of smart grid components. Therefore, smart grid stakeholders must follow these standards to ensure the safety and proper functioning of the modern power system. Coordination and corporation between national TSO and these international organisations are crucial for ensuring the reliability of distributed energy resources as a means of maintaining generation

[1] Interoperability [159] is the ability of two/more interconnected systems and components to exchange information and use that information to ensure their responses are secure and convenient to the user.

during fault events. The evolution of some international standards for DER interconnection is summarised in Fig. 6 [157].

5. Distributed generation in grid codes

Grid code regulations are being frequently modified to ensure stable and continuous power system operation in the presence of high DG technologies. At high DG penetration, self-disconnection of DG units under grid contingencies due to loss of main (LoM) protection in traditional grid codes can result in widespread power outages, cascading failures, and eventually, complete system blackout. Thus, the DG system must remain operating during voltage and frequency transients to avoid load shedding and possibly system collapse. In the following subsections modern grid codes incorporate dynamic regulatory requirements such as voltage and frequency ride-through are evaluated.

5.1. Low/high voltage ride through and reactive power control

Voltage sags and swells often occur throughout the year due to short circuits and rapid changes in sources and loads, which can last anywhere from half a cycle to a few seconds. When the sag percentage is too high, it is considered a fault event, and the DG inverters may be disconnected from the electric power system, as per IEEE 1547-2018 standards. Low voltage ride-through (LVRT) is the ability of DG to ride through a significant voltage dip at the PCC and remain connected to the grid for a certain time before the fault is cleared [167]. In such situations, it is necessary for both wind and solar PV converters to maintain synchronism and to exchange current and energise the power grid. Although most TSOs have identical grid codes for wind and PV systems, the LVRT strategies for these systems differ due to their various system characteristics [168].

Denmark, Ireland, the UK, and Germany have the most developed integrated wind power systems among the renewable dominant countries, while the UK, Spain, Germany, the United States, and China produce the most wind energy. Therefore, this section thoroughly investigates the voltage dip duration profile with LVRT requirements for large-scale wind farm integration in these countries using grid codes of Ireland EirGrid [169], Australia AEMC [170], UK National Grid [171], Germany and Netherlands TenneT [172], Denmark Energinet [173], India [174], Spain [175], China GB/T and NB/T [176], and USA ERCOT [177] as illustrated in Fig. 7.

As depicted, the LVRT standards are extremely stiff in countries with high DG penetration. As shown from the borderlines, most grid codes require wind turbines to remain connected for extremely low voltage down to 0 p.u. for a long time - ranging from 150 ms up to 400 ms from the instant of the fault. This is clear from Australian grid codes, which impose stringent voltage variation requirements with extremely low post-fault voltage recovery of 0.7 p.u.

In Germany, wind turbines are rewarded for their FRT capabilities, and the functionality is also mandatory for all new plants. The LVRT codes involve two lines; following a fault, the plant shall remain connected if the terminal voltage is above Germany 1. Between the first and the second line, the plant can be momentarily disconnected and reconnected back within 2 s, while below line 2, the plant can always be tripped. To support local voltage and limit the geographical low voltage area produced by the grid fault, wind farms must deliver a 2% reactive current to the grid for each percentage of voltage decrease at the PCC between 10% and 50% [178]. The active current can be reduced in such circumstances to increase the overall VA capability. This means the priority is given to the reactive current to maintain the PCC voltage.

In the Danish grid code, a wind farm with a rated capacity of more than 1.5 MW must comply with LVRT standards, and it must normally generate for a maximum of 5 s following a transient start-up phase. To ensure that the wind power plant assists in stabilising voltage within the design framework afforded by current wind power plant technology, wind power plants must provide 100% of the reactive current capacity if the PCC voltage drops by 50%. Similarly, in China wind power plants with a capacity larger than 1 GW shall inject reactive current with a response time of less than 75 ms and for a duration of longer than 550 ms until voltage recovery of 90% is achieved within 2 s [176].

In Ireland, wind generators are required to withstand voltage drops up to 85% of the nominal voltage for the longest timeframe of 625 ms during the fault and 10% after 3 s when the fault is cleared. However, as the power system of Ireland is isolated from large synchronous networks, priority is always given to active power. The reactive current response is only required within the remaining VA capability margin and is proportional to the PCC voltage sag. The provision of the reactive current must continue at least 500 ms until the voltage recovers up to 90% of nominal value and the wind generator active power should recover to 90% for faults cleared within 140 ms.

Year	1990's	2000's	2010's ~ 2020
Evolution of standards	IEEE 519-1992, ANSI C48.1-1995 UL 1741-1999	IEEE 929-2000 1EEE 1547-2003 UL 1741-2010 CSA C22.2 No.107.1-01	IEEE 1547a-2014 & 2018 & 2020 IEEE 2030.2 UL 1741 SA & CA Rule 21 & HI Rule 14H (USA) EN 50438:2013 & EN 50549:2019 (Europe) EREC G98 & G99 (Great Britain) VDE-AR-N 4105: 2011-08 & 2018_11 (Germany) CSA C22.2 No.107.1-16 & CSA C22.3 No.9 (Canada) CEI 0-21 (Italy)
Objections: Requirements for DER interconnection	Compliance with technical specifications of: Voltage range Frequency range Synchronisation Total harmonic distortion (THD)	Safety and protection: Power factor correction Anti-islanding Disconnection and reconnection Power quality improvement (i.e., THD, Flicker)	Power system support functions: Frequency and voltage regulation Frequency and voltage ride through Power curtailment Wider power factor adjustment Grid forming Black start

Fig. 6. Worldwide standard evaluation for DER interconnection [157].

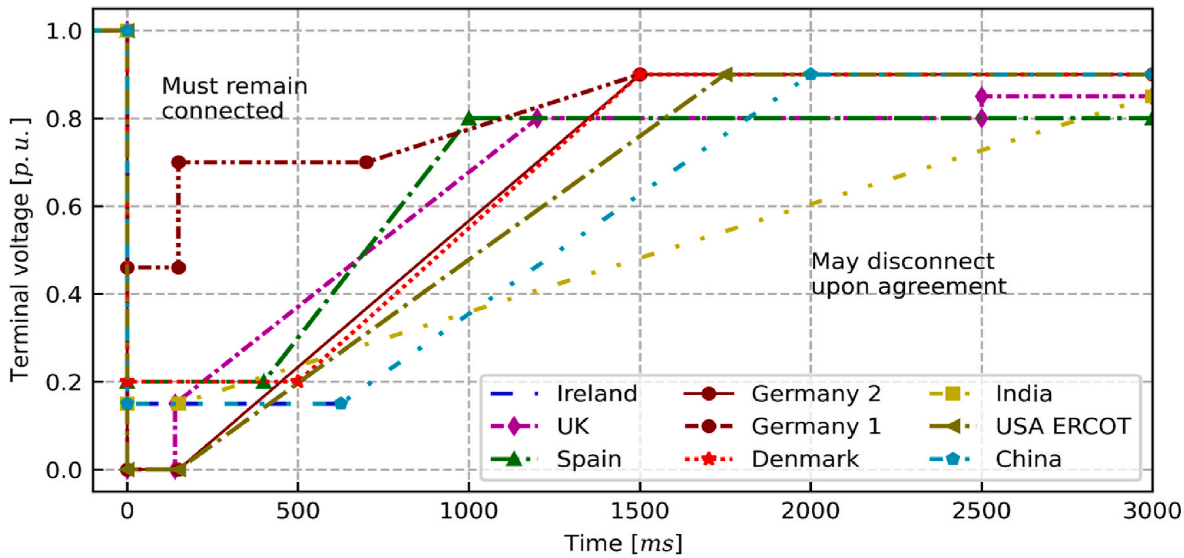

Fig. 7. Wind farm LVRT requirements in international grid codes.

The LVRT regulations in the UK are quite specific and strictly adhere to delicate points such as the division of wind turbines into onshore and offshore, different operative fault clearance times, two or three ended circuits, and various retained voltage types (i.e., 30%, 50%, and 85%) [168]. Similarly, active power is required in the UK to return to 90% of the wind turbine power before the fault within 500 ms following the fault event. Furthermore, similar active power recovery recommendations are established in German and Spanish grid codes following fault removal. Finally, other countries, such as Canada and the USA, employ different standards for the power system in each part of the country.

High voltage ride through (HVRT) capability, on the other hand, is seldomly addressed in the grid codes due to protection concerns. It is to be noted that the disturbance tolerance for voltage swells is substantially smaller than that for voltage sags hence LVRT occurs over a larger range than HVRT. As illustrated in Table 6, these requirements typically vary from 1.2 p.u. to 1.3 p.u. of the rated terminal voltage for durations ranging from 0.05 to 1.0 s for most grid codes.

Table 6
HVRT and reactive power requirements in various grid codes.

Regions	HVRT		Power factor (pf)		Q/P_N
	V_{max} (p.u.)	t_{max} (s)	Leading	Lagging	Reactive power (p.u. registered capacity)
Australia	1.3	0.05	0.93	0.93	0.395 (automatic)
Denmark A2 (11 kW–50 kW)	1.15	0.2	0.95 < *pf* < 1	0.95 < *pf* < 1	N/A
Denmark B (50 kW–1.5 MW)	1.15	0.2	0.995	0.995	0–0.1 supply 0–0.1 absorb
Denmark C (1.5 MW–25 MW)	1.2	0.1	0.975	0.975	0–0.228 supply 0–0.228 absorb
Denmark D (>25 MW)	1.2	0.1	0.95	0.95	0–0.33 supply 0–0.33 absorb
Germany[a]	1.26	0.1	0.95	0.925	0–0.33 supply 0–0.41 absorb
India	–	–	0.95	0.95	N/A
Ireland	1.1	–	0.95	0.95	0–0.33 supply 0–0.33 absorb
Spain	1.3	0.5	0.99	0.99	0–0.3 supply 0–0.3 absorb
UK	1.1	–	0.95	0.95	0–0.33 supply 0–0.33 absorb
China	–	–	0.95	0.95	N/A
USA ERCOT	1.175	0.2	0.95	0.95	N/A

[a] Depending on the network requirements, three variants are provided by German grid codes.

In the Danish grid code, generators rated more than 25 MW must have a reactive power capability of ±0.3 p.u. for active power requirements ranging from 0.2 to 0.8 p.u, and this has been gradually reduced from +/0.3 p.u to ±0.2 p.u. when active power goes from 0.8 p.u. to 1 p.u. This implies a lower reactive power need at high active power levels, which is a viable reactive power requirement in terms of wind generator cost and technical limitations.

Table 6 summarises the reactive power requirements defined in various grid codes for wind farms. These requirements are typically presented using *P-Q* capability indicating the active power output versus reactive power generation with respect to the plant registered capacity (P_N), as depicted in Fig. 7. As shown in Table 6 and Fig. 8, most of the countries specify a power factor in the range of 0.95 leading to 0.95 lagging, while delivering rated active power.

Although the reactive power profile is dependent on the generator's remaining *P-Q* capability margin, the power factor requirements should maintain the plant reactive power profile within the designated voltage ranges. The conditions may vary depending on the wind farm registered capacity in different countries. In the UK, for instance, linear reactive power consumption is required when the plant operates between 20% and 50% of its rated power, while there is no constraint on the lagging reactive power.

5.2. Techniques to support voltage ride-through requirements

Reactive power support requirements vary by country, such as the ratio of reactive current to voltage sag/swell and the maximum requisite reactive power [179]. Wind farms containing a large number of fixed speed induction generator (FSIG) wind turbines have limited LVRT capability due to their continuous reactive power consumption during transient periods. When the terminal voltage drops, the conventional FSIG requires adequate reactive power to re-magnetise the generator [180]. In such conditions, the rotor speed of FSIG is typically over-speed and moves to instability because of their electromagnetic torque reduction. Both Type 3 and Type 4 variable speed wind turbine (VSWT), on the other hand, can be designed to regulate reactive power output to support the period of low voltage and meet LVRT requirements [181]. Their current limited converter devices play a significant role in the dynamic response of VSWT to network fault events.

Ride-through functions can be accompanied by VAr and Watt sup-

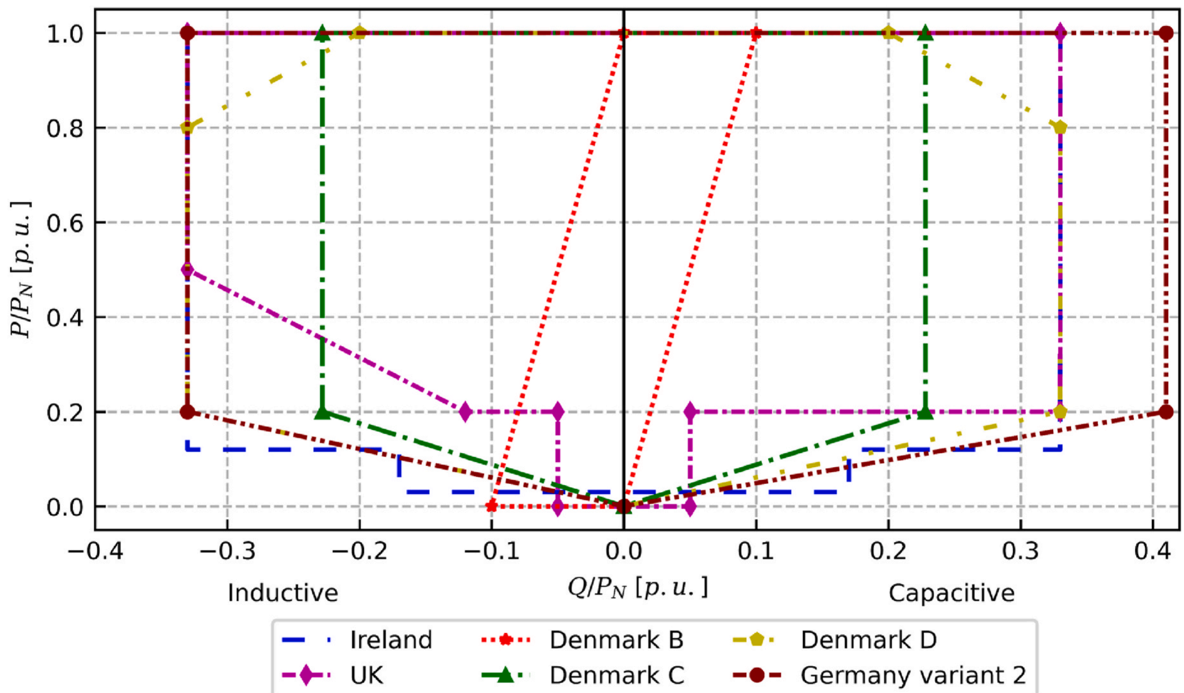

Fig. 8. Reactive power requirements in international grid codes.

port functions (Volt-Var ($Q-V$), Volt-Watt ($P-V$), $P-f$, dynamic reactive current support, dynamic Volt-Watt support). VAr related functions can be configured to have or not have precedence overactive power output. For each support function, several modes with $P-V$, $Q-V$ or $P-f$ curves can be defined for different periods (on-peak, off-peak) and operational conditions (islanded, grid-connected).

Techniques for LVRT in wind power systems can be divided into two categories: passive approaches that use auxiliary devices and active methods that utilize control schemes. Auxiliary devices include additional energy storage, reactive power compensators (i.e., Static var compensator), pitch drives for blade pitch angle regulation, crowbar for rotor voltage and current limitation, and an energy capacitor system comparable to crowbar connection [180].

As the development of additional devices raises system costs, active method control strategies such as non-crowbar method, robust controller in $\alpha-\beta$ stationary frame, virtual rotor resistance, proportional-integral (PI) controller with a resonant compensator, vector-based hysteresis current controller, and sliding mode control to manage rotor voltage and current can be utilized to support the LVRT requirements [182]. As the capability of control modes is limited compared to hardware modifications, hybrid solutions that combine the benefits of both strategies are becoming popular [183].

5.3. Frequency ride-through

Frequency ride-through regulations define the required response of DG units including wind turbines during high and low frequency conditions. These standards describe the active power response levels for a specified frequency range in the network, as well as the wind plant disconnection frequency thresholds. Fig. 9 shows frequency limitations in some international grid codes.

Some grid codes (i.e., EirGrid and TenneT) require wind turbines to supply the maximum possible active power for specific low-frequency ranges during under-frequency disturbances in contrast, during over-frequency events, the output power of the plant should be curtailed using a certain ramp rate. The DG plant is permitted to trip from service

Frequency (p.u.)	Australia (AEMC)	Denmark ENERGIENT (B, C, D)	India	Germany TenneT	Ireland (EirGrid)	UK (National Grid)	Spain	China (GB/T)	USA (ERCOT)
1.05				Disconnect		Disconnect			Disconnect
1.04	2 min	30 sec		10 sec		15 min			
1.03		30 min		30 min	60 min	90 min			
1.02	10 min								30 sec
1.01								5 min	9 min
1.004	Continuous	Continuous			Continuous	Continuous			
1			Continuous	Continuous			Continuous	Continuous	Continuous
0.99									
0.98	10 min							30 min	9 min
0.97				30 min	60 min				30 sec
0.96	2 min	30 min		20 min		90 min			2 sec
0.95				10 min			3 sec	WPP characteristics	
0.94		30 sec		10 sec	20 sec	20 sec			Disconnect
0.93						Disconnect			

Fig. 9. Frequency ride-through requirements in international grid codes.

when grid frequency exceeds the minimum and maximum frequency thresholds.

In the grid codes of Ireland, it is still not mandatory for the wind plants to operate below the optimal point to participate in frequency control. However, if they are operated below maximum power and equipped with frequency sensitive mode (FSM), it is mandatory to provide 60%–100% of their anticipated active power reserve within 5–15 s, respectively during large disturbances [169].

It is also necessary for the wind farms to receive a remote signal from the TSO to respond to active power curtailment signals. According to the TenneT onshore grid code, active power output should remain stable without constraint within the frequency range of 49–50.5 Hz. Nevertheless, when the frequency surpasses 50.2 Hz, active power is reduced based on frequency deviation from 50.2 Hz [184]. Active power curtailment is required for every frequency deviation at the rate of 40% if the frequency rises above 50.2 Hz–51.5 Hz. When the generator surpasses the maximum frequency threshold of 53.5 Hz and falls below the minimum frequency threshold of 46.5 Hz, it can disconnect. Similarly, frequency support is required in ERCOT only if the wind farms are curtailed to operate below MPPT [177].

The Spanish grid codes have the widest range of operating frequencies varying between 47.5 and 51.5 Hz and impose wind farms to have a maximum continuous operation time of at least 3 s when the frequency drops below 48 Hz. Finally, in Denmark, frequency support is essential within two 2 s of a disturbance, and the plant must curtail output power within 2–12% of its nominal power generation during high frequencies with an accuracy of ±10% [185].

It is essential to mention that the latest EirGrid grid codes set out technical obligations for wind turbines and include requirements for active power management in response to changes in system frequency. The technical specifications describe multiple operating modes and required behaviours during contingency events of varying severity. However, the grid code requirements are aligned to the primary operation reserve (POR) market rather than the new FFR market. This is representative of how the DS3 program has not yet qualified wind for the FFR market, and the grid code may be updated following the results of the Qualification Trial Process for wind under the DS3 program [186].

6. Discussion and recommendations

With the increasing integration of inverter interfaced renewable generation (i.e., wind, solar PV, energy storage) more and more traditional fossil fuel synchronous generators will be retired. This will result in a lack of frequency and voltage control due to a large reduction in system inertia and short circuit current level respectively. Several renewable generation technologies (e.g., energy storage, demand response, and enhanced inverter control algorithms) have been introduced to improve system inertia response and reactive power reserves in many power systems. Therefore, the first recommendation is that systematic programs must be deployed by network operators to manage these technologies within the available electricity market and infrastructure under a new regulatory framework.

For example, in 2021, Mitsubishi Electric Corporation (Japan) introduced new distributed energy resources management software called smarter grid solutions (SGS) for distributed energy resources operation management [187]. Next, all new renewable technologies should be mandated to offer voltage and frequency regulation and FRT to prevent trips and system collapse during disturbances.

The potential of several smart technologies to tackle grid stability challenges has been critically reviewed. A key recommendation is that GFM inverters should be prioritised as they are the most promising solution to mimic synchronous generators inertia response. Renewable enabling technologies can provide system stability services to increase grid operation security, voltage and frequency stability, and even black start capability if they are built in advance with intelligent coordination and control strategies such as GFM based inverters. Although GFL inverters can offer some ancillary services (i.e., FFR) they have limitations (i.e., PLL instability) in a weak grid. For a black start, synchronous condensers are highly recommended as a means of providing injection current to energise electromagnetic loads such as transformers and motor loads.

Many studies have been conducted to examine the dynamic stability operation of power systems with 100% inverter interfaced generation in small scale systems such as microgrids, but none have addressed the dynamic stability of the system at the grid scale. Therefore, although huge potential can be seen in the available VRE and smart technology solutions to maintain grid operation, security and reliability under 100% VRE, a proportion of synchronous generation are recommended to be online to run modern systems securely until sufficient operational experience is achieved.

In this review, a number of smart technologies to compensate reactive power and improve voltage stability in distribution networks are recommended, including synchronous compensator, static var compensator, static synchronous compensator (STATCOM), V-Q control of PV inverter, switched shunt capacitor, and energy storage applications. The capability of these smart technologies to provide dynamic reactive power response and voltage stability enhancement will bring confidence to move to a 100% VRE power system.

Given the variability and uncertainty of wind and solar PV, weather forecasts need to be more accurate to determine the real available wind and solar PV power in extreme events such as storms. Improved weather simulation data is recommended to diminish the uncertainty of VRE. Reduced forecast error will help grid operators to predict longer low wind events ensuring long term reliability and security of renewable generation.

Transmission system expansion and HVDC interconnected networks are recommended as potential solutions in terms of power balancing and system adequacy, especially in power systems with large renewable penetrations to overcome geographical weather spread. For instance, there is a huge potential between the European mainland, the UK and Ireland to harness both offshore and onshore wind and solar PV energy to meet Net Zero targets more efficiently if they were connected through HVDC.

Finally, the location and size of VRE and smart technologies have a significant impact on grid frequency response and bus voltage stability. National system operators are encouraged to pre-determine where and how to install new renewable technologies through detailed dynamic power system analysis.

7. Conclusion

This review examined and discussed the dynamic stability operation challenges facing modern power systems with extremely high VRE generation. This review focused on two issues: power system stability with high inverter-based resources and potential mitigation measures from various enabling technologies.

The first challenge is that, in response to Net Zero GHG emission targets, electric power systems are rapidly changing to include a high share of VRE generation such as wind, solar, EVs, smart appliances, and energy storage devices. However, wind and solar power are not without their own technical challenges. The main challenge associated with displacing large synchronous power plants with VRE is gradually losing system inertia and frequency and voltage control mechanisms. This will drastically change the dynamic behaviour of the power system, ensuring system stability and security a critical challenge.

The second challenge is that several enabling technologies (i.e., wind, solar PV, energy storage, EV, demand response, synchronous condenser, HVDC interconnectors, and enhanced converter control strategies) have been proposed in the literature, each with its own set of needs. The challenge is working out which technologies or mix of technologies are deployable at grid scale, and determining the requirements for these technologies so that they meet power system requirements. This review paper yielded five main key findings:

- Technical standards and grid code requirements for grid forming control operation have yet to be introduced. This is because industries lack clear requirements for the development of new grid-forming control generation to meet new power system paradigms.
- The installation of synchronous condensers will be a critical system requisite to ensure the security and strength of power systems with up to 100% inverter-based resources.
- There is no legislation requiring the installation of long-term hydrogen energy storage. Hydrogen is volatile energy that can explode. As a result, both local and international standards and regulations for such large-scale storage must be established, followed by social acceptance and awareness.
- Transmission expansion and international interconnections (e.g., HVDC) are key drivers to maintain the security of the supply of a power system with only a limited number of synchronous generators.
- It is unknown what levels of synchronous generation are required to maintain grid stability in power systems with up to 100% VRE. As a result, research into the amount of synchronous generation required to ensure grid stability and service levels while transitioning to Net Zero by 2050 is urgently needed.

In conclusion, this review highlights that power systems will undergo radical changes to approach the Net Zero targets by 2050. Thus, it is essential in the future analysis to consider the geo-political impact on security of supply and impacts of medium and low voltage DC distribution systems on integration of VRE penetration in future power systems.

CRediT authorship contribution statement

Faraedoon Ahmed: Conceptualization, Resources, Writing – original draft, Writing – review & editing, Visualization. **Aoife Foley:** Conceptualization, Writing – review & editing, Visualization, Supervision. **Seán McLoone:** Writing – review & editing, Visualization, Supervision. **Robert James Best:** Writing – review & editing, Visualization, Supervision. **Ché Cameron:** Writing – review & editing. **Dlzar Al Kez:** Conceptualization, Writing – review & editing, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Nomenclature

Abbreviation	*Description*
BESS	Battery energy storge system
CAES	Compressed air energy storage
CPP	Conventional Power Plant
DR	Demand response
DSM	Demand side management
DER	Distributed energy resources
EV	Electric vehicle
EFR	Enhanced frequency response
FRT	Fault ride through
FFR	Fast frequency response
FRT	Frequency ride through
G2V	Grid to vehicle
GFL	Grid following
GFM	Grid forming
GHG	Greenhouse gas
HES	Hydrogen energy storage
HVDC	High voltage direct current
IBR	Inverter based resources
I-SEM	Integrated single electricity market
LCOE	Levelised cost of energy
MG	Microgrid
PCC	Point of common coupling
PHES	Pumped hydro energy storage
PLL	Phase locked loop
PV	Photovoltaic
P–V	Active power-voltage
Q-V	Reactive power-voltage
RES	Renewable energy sources
RoCoF	Rate of change of frequency
SC	Synchronous condenser
SCL	Short current levels
SoC	State of charge
SRMC	Short running marginal cost
TSO	Transmission system operators
UC	Unit commitment
V2G	Vehicle to grid
VRE	Variable renewable energy

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